

Sieng Lyheng

Applied Mathematics & Statistics | Data Science Student

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Education

Institute of Technology of Cambodia Phnom Penh, Cambodia

B.Eng. Applied Mathematics & Statistics (Data Science) Expected 2027

Year 3 GPA: 4.00 / 4.00

Experience

Data Visualization for Library Resources & Attendance Aug 2024 – Sep 2024

ITC Library, Informal Internship Phnom Penh, Cambodia

- Analyzed book-borrowing and room-booking data from multiple departments using Python and Jupyter Notebook.
- Reported findings and insights to the Head of Library to support data-driven decision-making.

Projects

Data Engineering

Real-Time Weather API & Analytics System 2026

Python, Kafka, PySpark, PostgreSQL, Docker, Streamlit

- Built a real-time pipeline that collects WeatherAPI data through Kafka and processes it with PySpark Structured Streaming.
- Designed Bronze, Silver, and Gold layers in PostgreSQL and developed dashboards for monitoring and weather analytics.

Cybersecurity Attacks ETL & Analytics Pipeline 2026

Python, Pandas, SQL Server, PySpark, JDBC, Power BI

- Developed a batch ETL pipeline to clean and load approximately 40,000 cybersecurity records into Microsoft SQL Server.
- Used PySpark through JDBC for analysis and built a Power BI dashboard for attack trends, severity, protocols, and actions.

Data Analytics & Machine Learning

Loan Default Risk Analysis and Prediction 2026

Python, Pandas, Scikit-learn, Random Forest, Streamlit, Power BI

- Built a credit-risk workflow to analyze applicant data and predict loan-default likelihood.
- Applied feature engineering and Random Forest classification, then delivered a Streamlit app and Power BI dashboard.

Customer Retention & Churn Analysis Dashboard 2026

Power BI, Power Query, DAX, Excel

- Built a three-page dashboard to analyze churn, retention, customer profiles, and behavior across 5,630 customers.
- Cleaned data with Power Query and created DAX measures for churn, retention, revenue, and customer KPIs.

Projects Continued

Artificial Intelligence & Computer Vision

Sao Somleng Game — Khmer Voice Learning Game2026

Python, FastAPI, PyTorch, Wav2Vec2, React

- Developed a voice-controlled educational game that helps children practice Khmer character pronunciation.
- Fine-tuned Wav2Vec2 speech models and integrated continuous-play mode with performance tracking.

Automatic License Plate Recognition in Cambodia2025

Python, YOLOv8, PaddleOCR, EasyOCR, Roboflow, Flask

- Developed an AI system for detecting and recognizing Cambodian vehicle license plates.
- Used YOLOv8 for plate detection and compared PaddleOCR and EasyOCR for text-recognition performance.

Technical Skills

Programming	Python, R, SQL, Java, C, C++
Data Engineering	Kafka, PySpark, Structured Streaming, ETL, Medallion Architecture, Docker, SQLAlchemy, JDBC
Data Analytics	Pandas, Power Query, DAX, Data Cleaning, Exploratory Data Analysis, KPI Development
Machine Learning	Scikit-learn, TensorFlow/Keras, PyTorch, Random Forest, Feature Engineering, Classification, Deep Learning
Visualization	Power BI, Microsoft Excel, Matplotlib, Streamlit, Jupyter Notebook
Databases	Microsoft SQL Server, PostgreSQL, MySQL, MongoDB
Tools	Git, GitHub, VS Code, Docker Compose, pgAdmin, SQL Server Management Studio, MySQL Workbench
Web & APIs	FastAPI, Flask, REST APIs, React

Languages

Khmer (Native) · English (Intermediate)